S U S S A I

3D PLC Simulator

User Manual

Version 1.0 - Ladder (Clic02 and LogixPro style), 14 machines in 3D, Gamification

EARLY ACCESS (testing phase)



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Contents

1. Introduction 3

Who it is for . . . 3

Requirements . . . 4

2. Installation and first access 5

Free trial step by step . . . 5

Entry screen (internal login) . . . 5

Where your files are stored . . . 5

3. License, plans and computers 6

Plans and prices . . . 6

How to buy . . . 6

License rules . . . 6

Help > SUSSAI license . . . 6

4. Welcome screen: Normal Mode or Gamification 7

5. The work screen 8

3D camera controls . . . 8

"Show in scene" box . . . 8

6. The Ladder editor 9

How to draw . . . 9

Tools . . . 9

Addressing: Clic02 or LogixPro style . . . 9

Example: direct-on-line start with seal-in (scene 02 - Fan) . . . 10

7. Simulation, faults and maintenance 11

RUN and STOP . . . 11

Machine actions . . . 11

Faults for maintenance training . . . 11

8. The 14 machines 12

Scene 01 - Simple conveyor (sensor + box) . . . 12

Scene 02 - Industrial fan (direct-on-line start) . . . 13

Scene 03 - Mixing tank (pump + agitator) . . . 14

Scene 04 - Single-road traffic light (two-way) with pedestrian crossing . . . 16

Scene 05 - Two-road traffic light with pedestrians . . . 17

Scene 06 - 4-way intersection (4 traffic lights) with pedestrians . . . 18

Scene 07 - Parking barrier / gate . . . 20

Scene 08 - Bottle filling (silo + valve) . . . 21

Scene 09 - Building water tank (pumps A/B) . . . 23

Scene 10 - Turntable (sorts by size) . . . 24

Scene 11 - 4 cascading conveyors (accumulation) + operator . . . 26
Scene 12 - Hydraulic press - two-hand control and light curtain (NR-12) . . . 28
Scene 13 - Conveyor with sorting (advanced) . . . 29
Scene 14 - Passenger elevator (4 floors) . . . 31
9. Gamification Mode **33**
Available exercises . . . 33
10. Extra scenes (.cena files) **35**
11. Safety and applied standards **36**
Button and indicator colors (IEC 60204-1 / IEC 60073) . . . 36
Emergency stop button (NR-12) . . . 36
Two-hand control and light curtain (scene 12 - Press) . . . 36
Programming language . . . 36
12. Keyboard shortcuts **37**
13. Saving, opening and moving to another computer **37**
14. Troubleshooting **38**
15. Early Access: reporting problems and suggestions **38**
How to report a problem (takes 1 minute) . . . 38
Suggestions, questions and compliments . . . 38
What's new and new versions . . . 39
16. Contact and rights **39**

Language: SUSSAI is available in **English (US)**, **Spanish** and **Portuguese (Brazil)**. Choose the language in the language box on the top bar (PT-BR / EN-US / ES) of the program or with the links on the login screen; the whole program, the help and this manual follow the chosen language. Only the I/O addresses and tags (I0, Q1, B1, K1, T0, C0...) are the same in every language.

1. Introduction

SUSSAI - 3D PLC Simulator is a training environment for industrial automation classes. Students program a Programmable Logic Controller (PLC) in **Ladder** language (IEC 61131-3), in the style of **WEG Clic02** or **Allen-Bradley LogixPro** software, and watch the machine run in **3D**, reacting to the program in real time.

There are **14 industrial machines**, ordered from the easiest to the most complete, each with real PLC inputs and outputs, ready-made programs to study, challenges to complete and **faults** for maintenance training. **Gamification Mode** turns practice into guided exercises with a score from 0 to 10.

Who it is for

- Students and instructors of technical and industrial apprenticeship courses (electrician, electromechanics, mechatronics, automation).
- Beginners in electrical controls and PLCs who want to practice without risk before going to the workbench.
- Instructors who need standardized hands-on assessment (Gamification Mode).

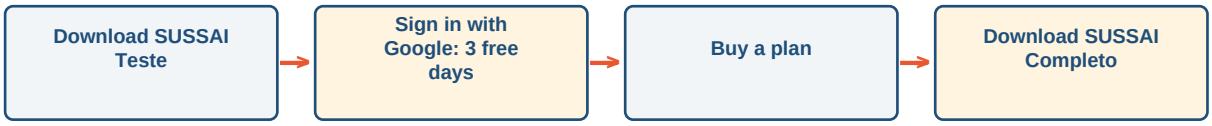
Requirements

- Windows 10 or 11, with Microsoft Edge (included in Windows) or Google Chrome.
- Google account (Gmail or school Google account) to sign in and activate the license.
- Internet only to sign in/activate. After that it works offline until the end of the plan.
- Recommended: screen of 1366 x 768 or larger. On slower computers, use Low image quality.

2. Installation and first access

SUSSAI comes in **two versions**. Neither needs installation: the program is a single file.

Version	What it is for	Where to get it
SUSSAI Teste (trial - SUSSAI_Teste.exe)	Free **3-day** trial with scenes 01, 02 and 03 (normal mode and gamification).	"Free download" button at sussai-software.github.io
SUSSAI Completo (full version - SUSSAI.exe)	All **14 scenes** and all exercises. Opens only with a paid license.	After purchase, SUSSAI Teste itself shows the "Download SUSSAI Completo" button.



Free trial step by step

- Download the file **SUSSAI_Teste_Windows.zip** from the website and unzip it (right-click > Extract All).
- Double-click **SUSSAI_Teste.exe**. It opens in its own window using Edge or Chrome.
- On the **"Bem-vindo ao SUSSAI"** (Welcome to SUSSAI) screen, type your **name** and your **mobile/WhatsApp number with area code**.
- Click **"Sign in with Google"** and choose your account. Done: **3 free days** unlocked right away.
- The free trial is given **once per Google account and once per computer**.

Windows warning: because it is a new program, Windows may show "Windows protected your PC". Click **"More info" > "Run anyway"**. If a blocked pop-up window warning appears, allow the Google sign-in window.

Entry screen (internal login)

After the license, SUSSAI shows the entry screen with the machines running in the background. Use **user: visitante** and **password: 3015** (or your own user name, to keep each student's scores separate on the same computer).

Where your files are stored

Saved circuits, scores and the license are stored on the computer itself, in the folder **%LOCALAPPDATA%\SUSSAI**. Deleting this folder resets SUSSAI on that computer (circuits and scores are lost).

3. License, plans and computers

Plans and prices

Plan	Duration	Computers at the same time	Price
Monthly	1 month	2	R$ 30,00
Quarterly	3 months	2	R$ 60,00
Semiannual	6 months	2	R$ 110,00
Lifetime	never expires	2	R$ 200.00 (one-time payment)
Group (schools, labs, companies)	1 year	up to 20	R$ 1,000.00 per year

Reference prices for this edition of the manual. Up-to-date prices are always at **https://sussai-software.github.io**.

How to buy

- Inside the program, click **"I want to buy"** (shown when the trial opens and in Help > License), choose the plan and send.
- Or write to **contato.sussai@gmail.com** with the plan and the Google e-mail you use in SUSSAI.
- After payment, the license is released and SUSSAI updates by itself (or click **"Já comprei - conferir agora"** / I already bought - check now).

License rules

- The license is tied to the **Google account**. Always use the same account.
- The term starts counting when the license is released.
- When the plan ends, SUSSAI **stops opening, even offline**. Your circuits and scores remain saved.
- During the **first 7 days** after release SUSSAI checks the license once a day; after that, every 30 days (when there is internet).
- **Changed computers?** Help > SUSSAI license > "Disconnect this computer". The seat becomes free for the new computer.
- **Do not set the Windows date back**: SUSSAI locks until the date is corrected.
- Withdrawal: within 7 days after release, with a refund (Brazilian Consumer Protection Code, art. 49). The license is then blocked.

Help > SUSSAI license

Shows the Google account, the plan, the expiration date, the code and this computer's ID. Buttons: **Check now** (looks for a renewal), **Buy / renew** and **Disconnect this computer**.

4. Welcome screen: Normal Mode or Gamification

Mode	What it does	Best for
1 - Normal Mode	You program in Ladder freely. The diagram opens blank; the Program menu has the ready-made programs (COMPLETE) and the CHALLENGES (one contact missing / one link missing).	Classes, free study, building and testing logic.
2 - Gamification Mode	Exercises with a ready circuit and a sequence of steps: operate the machine, answer, find a hidden fault and fix the program. At the end the **score from 0 to 10** is shown.	Hands-on assessment, review and healthy competition.

To change mode: **File > Welcome screen**. To return to the login: **File > Exit**.

5. The work screen

The screen has the **3D machine**, the **Ladder diagram**, the **inputs and outputs** panel and the status bar. All commands are in the menu bar at the top:

Menu	What it contains
File	New, ready-made programs, My circuits, open/download .json file, save (Ctrl+S), export PNG image, welcome screen, exit.
Edit	Undo/redo, insert/delete rungs, advanced instructions, Names (aliases for inputs, outputs and memories).
Scene	Choose the machine (01 = easiest) and **Adicionar cena nova (.cena)** (Add new scene).
Program	Ready-made programs for the current machine: complete, step by step, with an intentional error, and challenges.
Simulation	PLC RUN/STOP, machine actions (place a box, call a pedestrian...) and **faults** for training.
View	What appears on screen, cameras, **quality (Low / Medium / High)**, two screens, Clic02 or LogixPro addressing style, **language** (PT-BR / EN-US / ES).
Tools	I/O table, memories, maintenance, event log, names.
Help	License, tutorial, machine help, button colors (IEC 60204-1), shortcuts, Report a problem, Send a suggestion, What's new and About.

3D camera controls

Mouse action	Result
Drag with the left button	Rotate the view
Drag with the right button	Pan the view
Mouse wheel	Zoom in / out
Double-click an object	Zoom to the object
Click the 3D panel buttons	Press the button (as on the real machine). The emergency stop latches; click again to release it.

"Show in scene" box

In the lower right corner of the scene is the **Show in scene** box (click ■ to collapse it):

Option	What it does
Addresses (I / Q)	Shows a tag with the PLC address on each button, lamp, motor, sensor and valve (green = input, orange = output). It follows the chosen style: I01/Q01 (Clic02) or I:1/0 and O:2/0 (LogixPro). Uncheck it to keep the scene clean.
Descriptions	Shows the names and explanations of the parts.
Pedestrian lights	Only in the 2-way and 4-way traffic light scenes: starts OFF (cars only). Check it to install the pedestrian lights and push buttons and increase the difficulty.
Low / Medium / High	Low: light, no shadows (PC without a graphics card). Medium: balanced. High: game mode (soft shadows, ambient occlusion, lamp glow, anti-aliasing).

Slow computer? Choose **Low** in the Show in scene box (or in View > 3D image quality). SUSSAI also lowers the quality by itself when the image stutters. Hover the mouse over any lamp or button to see its name and address.

Image stuttering even with a good graphics card? Check **Performance** in the Show in scene box and look at the card name. If it shows Intel or AMD (integrated) – SUSSAI also warns "integrated card" – open Windows > Settings > System > Display > Graphics, choose **Microsoft Edge** > Options > **High performance** and open SUSSAI again.

6. The Ladder editor

Ladder is drawn in rows (rungs) between two power rails. Each rung connects contacts (conditions) to an output (coil, timer or counter).

How to draw

- Choose the element on the toolbar (or use the shortcut key).
- Click with the **left button** at the position in the diagram to insert it.
- To delete: click the element once and press **Delete**.
- **Double-click** opens the properties window (address, timer preset, counter value), in the Clic02 style.
- Use the **vertical link** to build parallel branches (for example, the seal-in contact).

Tools

Tool (Clic02)	LogixPro equivalent	Key	Function
Select	Select	Esc	Selects and edits elements
NO contact	XIC	1	Normally open contact: conducts when the bit = 1
NC contact	XIO	2	Normally closed contact: conducts when the bit = 0
Edge (rising) (P)	XIC (rising) (ONS)	-	Conducts for 1 scan on the rising edge of the bit
Edge (falling) (N)	XIC (falling)	-	Conducts for 1 scan on the falling edge of the bit
Line -	Wire -	8	Horizontal line (click and drag to draw)
Link |	Branch |	9	Vertical link (parallel). Click near the border between columns
Coil ( )	OTE	3	Output coil: bit = rung condition
Set (S)	OTL	-	Turns on and holds (retentive). In this course, prefer a regular coil with a seal-in contact.
Reset (R)	OTU	-	Turns off the retentive bit. In this course, prefer a regular coil with a seal-in contact.
TON	TON	6	On-delay timer
TOF	TOF	-	Off-delay timer
CTU	CTU	7	Up counter
CTD	CTD	-	Down counter
RES	RES	-	Resets timer/counter
Eraser	Delete	0	Deletes the clicked element

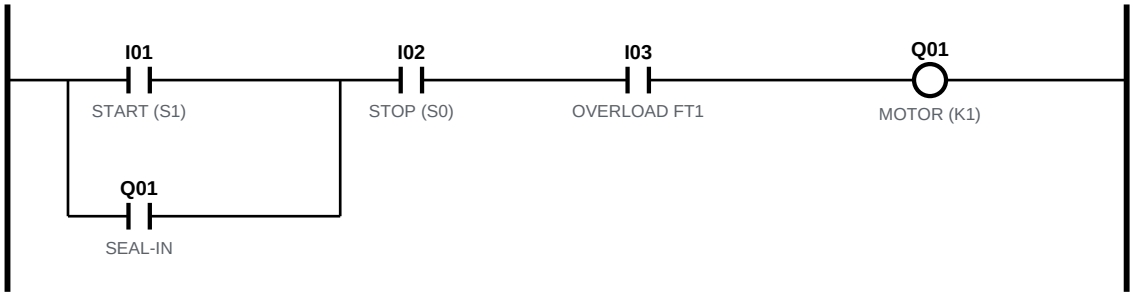
Addressing: Clic02 or LogixPro style

Choose it in **View > PLC style**. The program is the same; only the way addresses are written changes:

Element	Clic02 style	LogixPro style
1st input	I01	I:1/0
2nd input	I02	I:1/1
1st output	Q01	O:2/0
Memory (internal bit)	M01	B3:0/0
Timer / done bit	T01 / T01	T4:0 / T4:0/DN
Counter / done bit	C01 / C01	C5:0 / C5:0/DN

Extra addresses: besides the machine inputs and outputs, there are I01 to I100, Q01 to Q100, M01 to M256, T01 to T100 and C01 to C100. An extra input becomes a **virtual switch** and an extra output becomes a **virtual lamp** on the panel.

Example: direct-on-line start with seal-in (scene 02 - Fan)



START (LIGA) energizes the motor coil; the **seal-in** contact (from the motor itself) in parallel keeps the motor running after the button is released. **STOP** (DESLIGA) and the **overload relay FT1** are **NC in the field**: at rest the input is 1, so in the program we use a **NO contact** for them, in series. Pressing STOP or tripping the overload opens the rung and the motor stops.

Safety rule: stop buttons, overload relays and emergency stops are always NC in the field (fail-safe: broken wire = machine stops). Using an NC contact in the program for a signal that is already NC in the field inverts the logic and is **unsafe**.

Prefer a coil with a seal-in contact over SET/RESET (retentive). All ready-made SUSSAI programs use only regular coils with seal-in.

7. Simulation, faults and maintenance

RUN and STOP

F5 puts the PLC in RUN (executing the program) and **F6** in STOP. In RUN the energized rungs turn green and the active inputs/outputs light up in the I/O panel. You can force inputs and outputs for testing (Tools > Inputs and Outputs).

Machine actions

The **Simulation** menu holds the actions of each scene: place a box, call a pedestrian, change the amount of cars, open the tap, etc. The 3D panel buttons can also be clicked directly.

Faults for maintenance training

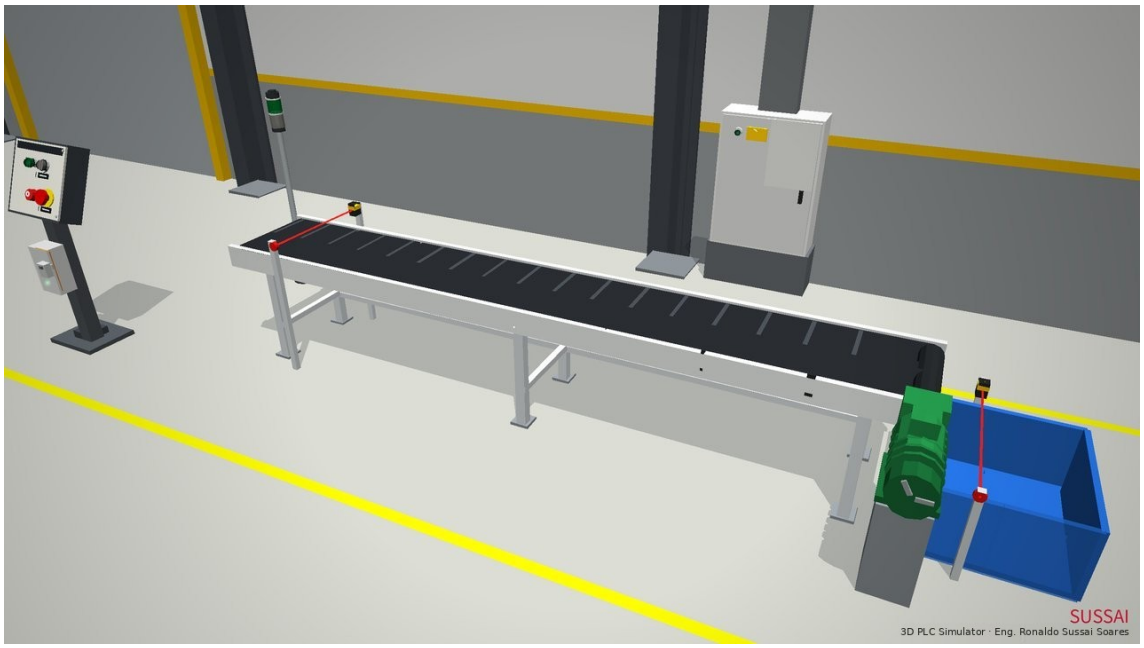
In **Simulation > Faults and training** the instructor (or the student) activates real faults: motor overload (the overload relay trips and smoke comes out), dirty or stuck sensor, broken wire, welded contact, burnt-out lamp, two-hand control bypass. The student must observe the machine, read the inputs in the I/O panel and explain the fault. The fault list of each machine is in chapter 8.

Overload reset: after an overload, the machine only restarts after the overload relay is reset and a new START command is given. It never restarts by itself (Brazilian safety standard NR-12).

8. The 14 machines

The machines are in order of difficulty. Every machine with a motor has an **overload relay** wired to a PLC input and every panel has an **emergency stop button** (NR-12). The tables below were generated from SUSSAI itself.

Scene 01 - Simple conveyor (sensor + box)



A conveyor with two sensors: B1 at the start of the belt and B2 at the mouth of the bin. A box at B1 starts the conveyor; when it falls into the bin and passes B2, the conveyor stops. One box at a time.

Also available in **SUSSAI Teste** (free trial).

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	B1	SENSOR ENTRADA (entry sensor)	Optical sensor B1 - box at the start of the conveyor
I04 / I:1/3	B2	SENSOR RECIPIENTE (bin sensor)	Optical barrier B2 at the mouth of the bin - 1 while the box is falling into it
I05 / I:1/4	FT1	TÉRMICO M1 (overload M1)	Overload relay FT1 of conveyor motor M1 - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I06 / I:1/5	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	M1	MOTOR ESTEIRA (conveyor motor)	Contactor K1 - conveyor motor M1
Q02 / O:2/1	H1	LÂMPADA VERDE (green lamp)	Green indicator - system on, ready to operate
Q03 / O:2/2	H2	LÂMPADA BRANCA (white lamp)	White indicator - conveyor moving (motor running)

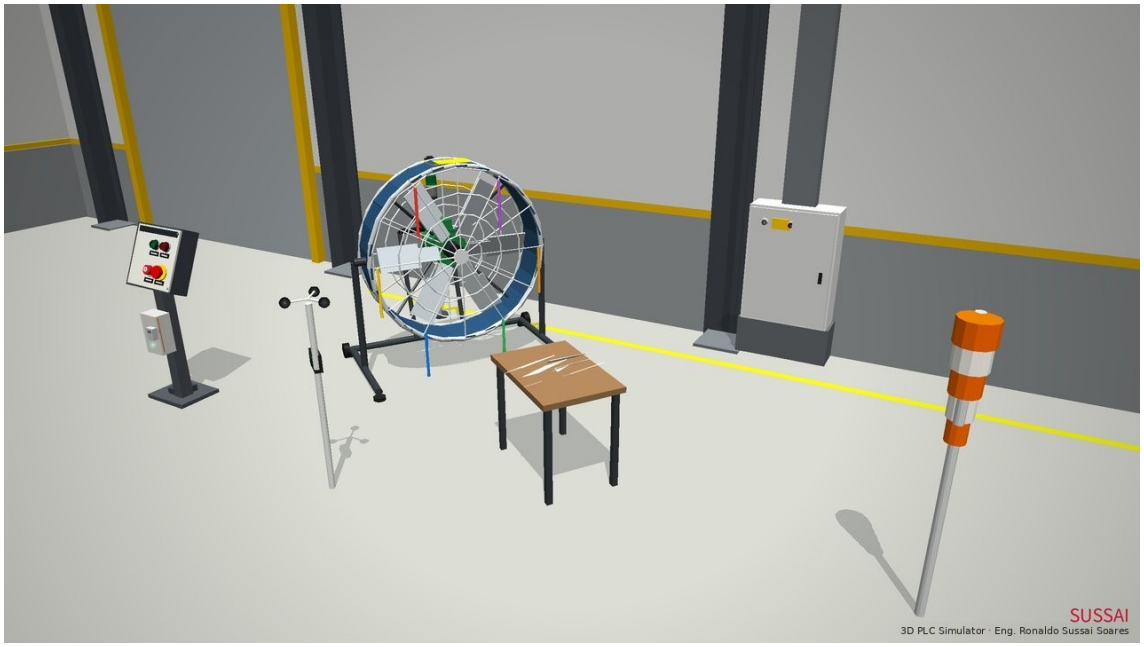
Ready-made programs (Program menu)

- 1 - Box on the sensor starts the conveyor (simplest)
- 2 - Start/Stop with seal-in
- 3 - Automatic with START/STOP
- 4 - Several boxes (CTU/CTD counter)

Training faults (Simulation menu)

- Overload on conveyor motor M1 (overload relay FT1)
- Sensor B1 with dirty lens (does not detect)
- Sensor B2 stuck at 1
- Motor M1 burnt out
- Slipping belt
- Green lamp H1 burnt out

Scene 02 - Industrial fan (direct-on-line start)



Build the direct-on-line start with seal-in, indicator lamps and overload relay. To do it for a SCORE, use Gamification Mode.

Also available in **SUSSAI Teste** (free trial).

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	FT1	TÉRMICO MOTOR (motor overload)	Overload relay FT1 of the fan motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I04 / I:1/3	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	K1	MOTOR VENTILADOR (fan motor)	Contactor K1 - fan motor (direct-on-line start)
Q02 / O:2/1	H1	LÂMPADA BRANCA (white lamp)	White indicator - fan ON (machine moving)
Q03 / O:2/2	H2	LÂMPADA VERDE (green lamp)	Green indicator - fan OFF, ready to operate
Q04 / O:2/3	H3	LÂMPADA VERMELHA (red lamp)	Red indicator - protective stop: overload relay tripped

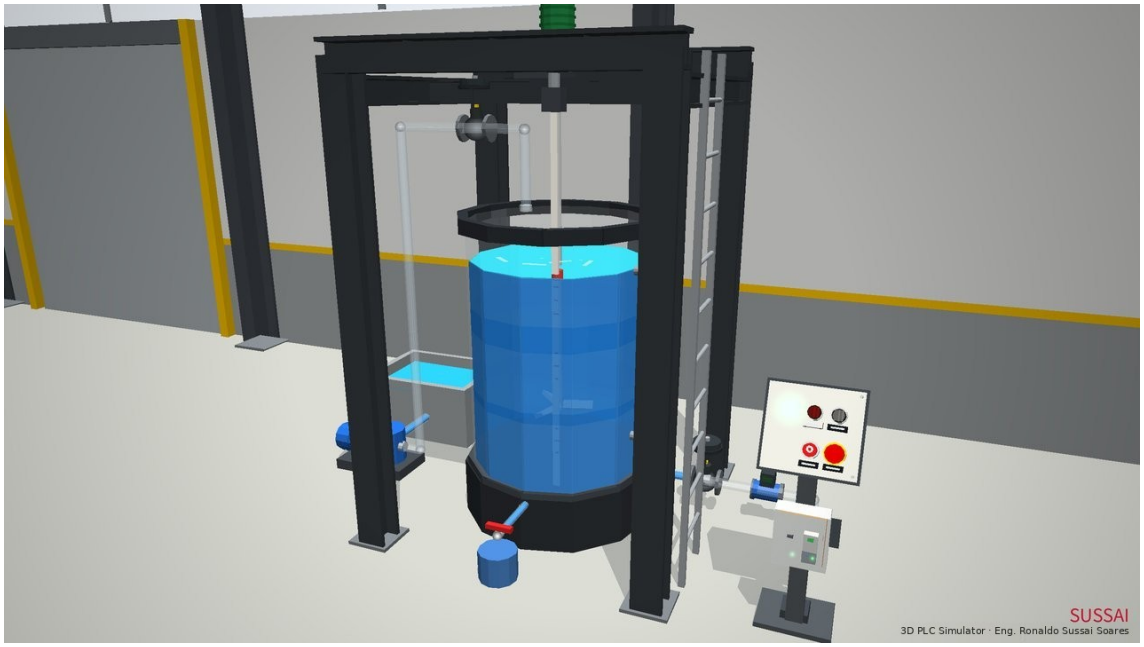
Ready-made programs (Program menu)

- Exercise - build the direct-on-line start yourself (blank)
- Answer key - direct-on-line start with seal-in, indicators and overload
- With an ERROR: no overload relay (the motor burns out)

Training faults (Simulation menu)

- Overload on the fan motor (overload relay FT1)
- Broken wire on the STOP button

Scene 03 - Mixing tank (pump + agitator)



The pump fills the tank, the agitator (propeller) stirs the liquid - after 5 s of mixing it changes color - and the finished product leaves through the outlet valve.

Also available in **SUSSAI Teste** (free trial).

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	LSL	NÍVEL BAIXO (low level)	Level sensor LSL at 20 % - reads 1 with liquid above it
I04 / I:1/3	LSH	NÍVEL ALTO (high level)	Level sensor LSH at 85 % - reads 1 with liquid above it
I05 / I:1/4	FT1	TÉRMICO BOMBA (pump overload)	Overload relay FT1 of the pump motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I06 / I:1/5	FT2	TÉRMICO AGITADOR (agitator overload)	Overload relay FT2 of the agitator motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I07 / I:1/6	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	K1	BOMBA (pump)	Contactor K1 - filling pump
Q02 / O:2/1	Y1	VÁLVULA SAÍDA (outlet valve)	Solenoid Y1 - outlet valve (product discharge)
Q03 / O:2/2	H1	LÂMPADA VERDE (green lamp)	Green indicator - system on
Q04 / O:2/3	H2	LÂMPADA VERMELHA (red lamp)	Red indicator - alarm (low level / overload)
Q05 / O:2/4	K2	AGITADOR (agitator)	Contactor K2 - agitator motor M2 (propeller)
Q06 / O:2/5	H3	LÂMPADA BRANCA (white lamp)	White indicator - product ready (mixed), discharging (information)

Ready-made programs (Program menu)

- 1 - Pump on/off (manual)
- 2 - Automatic level control
- 3 - Fill and empty (cycle)
- 4 - Mixer: fill -> stir 5 s -> empty

Training faults (Simulation menu)

- Overload on the pump motor (overload relay FT1)
- Overload on the agitator motor (overload relay FT2)
- HIGH level sensor (LSH) burnt out
- LOW level sensor (LSL) stuck at 1
- Pump motor burnt out
- Valve Y1 stuck closed
- Agitator propeller loose on the shaft

Scene 04 - Single-road traffic light (two-way) with pedestrian crossing



Two-way street with a mid-block crosswalk. It is an ON-DEMAND light: cars have GREEN all the time and the light only closes when a pedestrian presses the push button. After the minimum green, both car lights go to yellow together, close, and the walking figure turns green.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	BP	BOTOEIRA PEDESTRE (pedestrian button)	Pedestrian push button (on the crossing poles) - NO, BLUE button: 1 while the pedestrian presses it
I04 / I:1/3	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	VD	CARROS VERDE (cars green)	Car traffic light - green lamp
Q02 / O:2/1	AM	CARROS AMARELO (cars yellow)	Car traffic light - yellow lamp
Q03 / O:2/2	VM	CARROS VERMELHO (cars red)	Car traffic light - red lamp
Q04 / O:2/3	PV	PEDESTRE VERDE (pedestrian green)	Pedestrian light - green walking figure (may cross)
Q05 / O:2/4	PR	PEDESTRE VERMELHO (pedestrian red)	Pedestrian light - red hand (stop). Flashing = about to close
Q06 / O:2/5	AG	AGUARDE (WAIT)	"AGUARDE" (WAIT) lamp on the push button - crossing request registered

Ready-made programs (Program menu)

- 1 - Crossing with push button (complete)
- 2 - Step 1: car light only

Training faults (Simulation menu)

- Faulty pedestrian push button (does not actuate)
- Car RED lamp burnt out
- Pedestrian RED lamp burnt out

Scene 05 - Two-road traffic light with pedestrians



Intersection of two one-way streets with two traffic lights and a pedestrian crossing with push button. If both roads open at the same time, there is a collision!

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	SN	MODO NOTURNO (night mode)	Selector switch - night mode (flashing yellow)
I04 / I:1/3	BP	BOTOEIRA PEDESTRE (pedestrian button)	Pedestrian push button (on the crossing poles) - NO, BLUE button: 1 while the pedestrian presses it
I05 / I:1/4	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	A-VD	VIA A VERDE (road A green)	Road A traffic light - green lamp
Q02 / O:2/1	A-AM	VIA A AMARELO (road A yellow)	Road A traffic light - yellow lamp
Q03 / O:2/2	A-VM	VIA A VERMELHO (road A red)	Road A traffic light - red lamp
Q04 / O:2/3	B-VD	VIA B VERDE (road B green)	Road B traffic light - green lamp

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
Q05 / O:2/4	B-AM	VIA B AMARELO (road B yellow)	Road B traffic light - yellow lamp
Q06 / O:2/5	B-VM	VIA B VERMELHO (road B red)	Road B traffic light - red lamp
Q07 / O:2/6	P-VD	PEDESTRE VERDE (pedestrian green)	Pedestrian lights (all crosswalks) - green walking figure
Q08 / O:2/7	P-VM	PEDESTRE VERMELHO (pedestrian red)	Pedestrian lights (all crosswalks) - red hand. Flashing = about to close

Ready-made programs (Program menu)

- 1 - Traffic light cycle (timers)
- 2 - With night mode (flashing yellow)
- 3 - With pedestrian crossing (push button)

Training faults (Simulation menu)

- Road A RED lamp burnt out
- Road B YELLOW lamp burnt out
- Night mode switch with welded contact
- Faulty pedestrian push button (does not actuate)

Scene 06 - 4-way intersection (4 traffic lights) with pedestrians



Two two-way avenues: 4 car traffic lights (A1/A2 wired together on road A, B1/B2 on road B) and pedestrian lights with push buttons on the 4 corners.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	SN	MODO NOTURNO (night mode)	Selector switch - night mode (flashing yellow)
I04 / I:1/3	BP	BOTOEIRA PEDESTRE (pedestrian button)	Pedestrian push button (on the crossing poles) - NO, BLUE button: 1 while the pedestrian presses it
I05 / I:1/4	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	A-VD	VIA A VERDE (road A green)	Traffic lights A1 and A2 (road A, wired in parallel) - green lamp
Q02 / O:2/1	A-AM	VIA A AMARELO (road A yellow)	Traffic lights A1 and A2 (road A, wired in parallel) - yellow lamp
Q03 / O:2/2	A-VM	VIA A VERMELHO (road A red)	Traffic lights A1 and A2 (road A, wired in parallel) - red lamp
Q04 / O:2/3	B-VD	VIA B VERDE (road B green)	Traffic lights B1 and B2 (road B, wired in parallel) - green lamp
Q05 / O:2/4	B-AM	VIA B AMARELO (road B yellow)	Traffic lights B1 and B2 (road B, wired in parallel) - yellow lamp
Q06 / O:2/5	B-VM	VIA B VERMELHO (road B red)	Traffic lights B1 and B2 (road B, wired in parallel) - red lamp
Q07 / O:2/6	P-VD	PEDESTRE VERDE (pedestrian green)	Pedestrian lights (all crosswalks) - green walking figure
Q08 / O:2/7	P-VM	PEDESTRE VERMELHO (pedestrian red)	Pedestrian lights (all crosswalks) - red hand. Flashing = about to close

Ready-made programs (Program menu)

- 1 - Traffic light cycle (timers)
- 2 - With night mode (flashing yellow)
- 3 - With pedestrian crossing (push button)

Training faults (Simulation menu)

- Road A RED lamp burnt out
- Road B YELLOW lamp burnt out
- Night mode switch with welded contact
- Faulty pedestrian push button (does not actuate)

Scene 07 - Parking barrier / gate



ARTICULATED barrier (two arms) on an island in the middle of the road, high enough for a truck to pass. The car arrives, the loop detects it, the barrier opens. With a car underneath it does not lower; after the car passes, it closes. Cars arrive at random intervals.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	B1	LAÇO CHEGADA (arrival loop)	Loop detector B1 - car arrived at the barrier
I04 / I:1/3	B2	BARREIRA SEGURANÇA (safety barrier)	Photocells B2 - car UNDER the barrier
I05 / I:1/4	B3	LAÇO SAÍDA (exit loop)	Loop detector B3 - car passed the barrier
I06 / I:1/5	FA	FIM DE CURSO ABERTA (limit switch open)	Limit switch FA - barrier fully open
I07 / I:1/6	FF	FIM DE CURSO FECHADA (limit switch closed)	Limit switch FF - barrier fully closed
I08 / I:1/7	FT1	TÉRMICO CANCELA (barrier overload)	Overload relay FT1 of the barrier motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I09 / I:1/8	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	K1	ABRIR (open)	Contactor K1 - barrier motor up (opens)
Q02 / O:2/1	K2	FECHAR (close)	Contactor K2 - barrier motor down (closes)
Q03 / O:2/2	H1	SINAL VERDE (green signal)	Green light - you may pass

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
Q04 / O:2/3	H2	SINAL VERMELHO (red signal)	Red light - wait
Q05 / O:2/4	H3	GIROFLEX (beacon)	Amber warning beacon - barrier moving

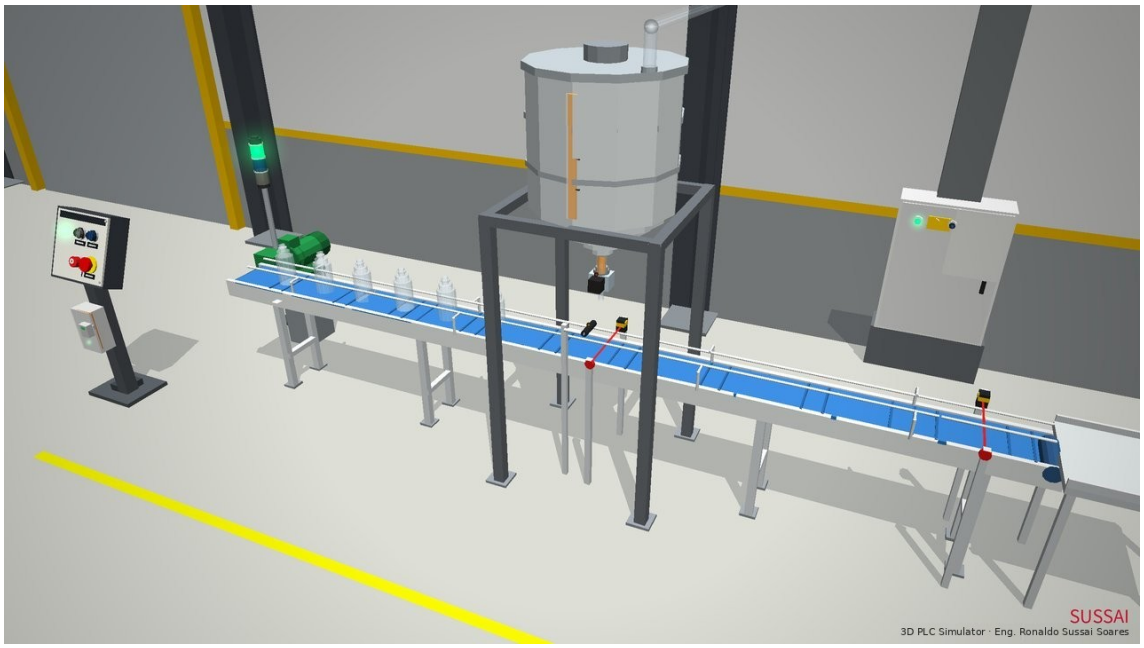
Ready-made programs (Program menu)

- 1 - Automatic barrier with sensors
- 2 - Barrier with timer (closes after 5 s)

Training faults (Simulation menu)

- Overload on the barrier motor (overload relay FT1)
- Safety barrier B2 burnt out
- Faulty arrival loop B1
- Limit switch FA (open) out of adjustment
- Barrier motor burnt out

Scene 08 - Bottle filling (silo + valve)



The bottle stops under the silo, the valve opens and fills it; the full sensor closes the valve and the bottle moves on. Then comes the next one.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	B1	GARRAFA NA POSIÇÃO (bottle in position)	Optical sensor B1 - bottle under the filling nozzle
I04 / I:1/3	B2	GARRAFA CHEIA (bottle full)	Capacitive sensor B2 - at the filling height: reads 1 when the product reaches the level

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I05 / I:1/4	B3	FIM DA ESTEIRA (end of conveyor)	Optical sensor B3 - bottle at the end of the conveyor (counting)
I06 / I:1/5	FT1	TÉRMICO ESTEIRA (conveyor overload)	Overload relay FT1 of the conveyor motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I07 / I:1/6	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	K1	ESTEIRA (conveyor)	Contactor K1 - conveyor motor
Q02 / O:2/1	Y1	VÁLVULA DO SILO (silo valve)	Solenoid Y1 - filling valve (opens the silo outlet)
Q03 / O:2/2	H1	LÂMPADA VERDE (green lamp)	Green indicator - system on
Q04 / O:2/3	H2	LÂMPADA BRANCA (white lamp)	White indicator - filling (valve operating)
Q05 / O:2/4	H3	LÂMPADA AZUL (blue lamp)	Blue indicator - batch complete: operator ACTION (remove the batch and press START)

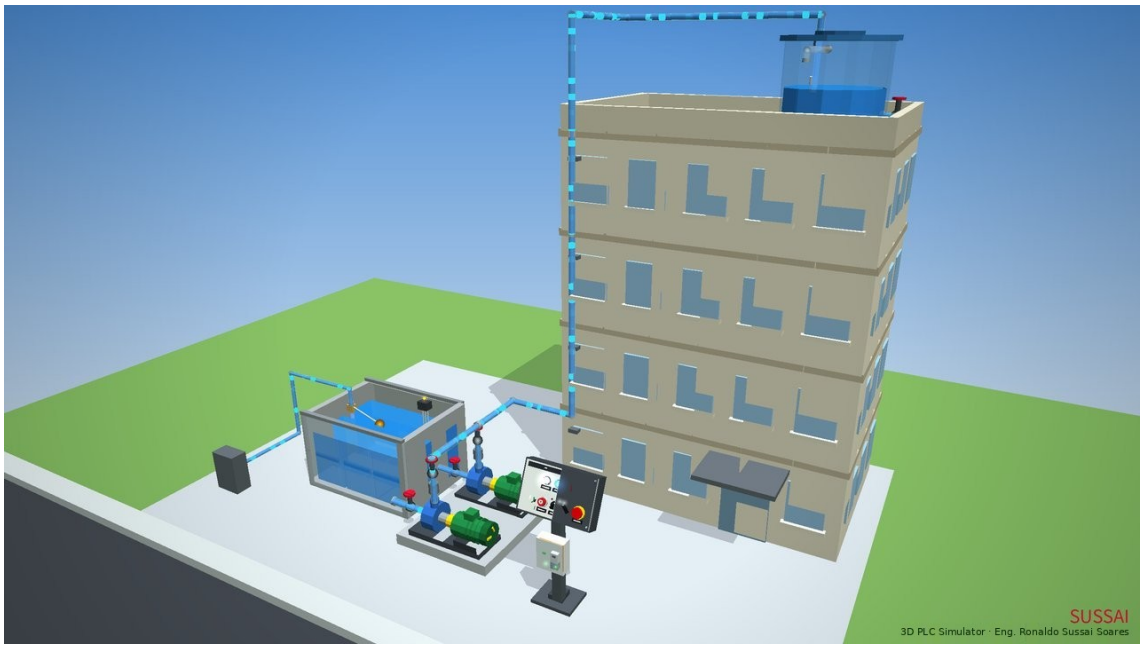
Ready-made programs (Program menu)

- 1 - Automatic filling (position sensor + full sensor)
- 2 - Batch of 6 bottles (counter)
- 3 - Step 1: stop the bottle under the silo

Training faults (Simulation menu)

- Overload on the conveyor motor (overload relay FT1)
- Faulty bottle FULL sensor (B2)
- Position sensor B1 out of place
- Valve Y1 stuck closed
- Clogged filling nozzle
- Conveyor motor burnt out

Scene 09 - Building water tank (pumps A/B)



The ground-floor cistern sends water to the top-floor tank through pump A or B. Automatic by the float switch or manual by the buttons.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	SA	MAN / AUTO	Selector switch: 0 = MANUAL (START/STOP buttons), 1 = AUTOMATIC (float switch)
I04 / I:1/3	SB	BOMBA A / B (pump A / B)	Selector switch: 0 = PUMP A, 1 = PUMP B
I05 / I:1/4	LS	BOIA CAIXA SUPERIOR (upper tank float)	Electric float switch of the top-floor tank: 1 = low level (asks for water), 0 = tank full
I06 / I:1/5	LC	ELETRODO CISTERNA (cistern electrode)	Cistern level electrode: 1 = there is water, 0 = minimum level (protects the pumps)
I07 / I:1/6	FT1	TÉRMICO BOMBA A (pump A overload)	Overload relay FT1 of pump motor A - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I08 / I:1/7	FT2	TÉRMICO BOMBA B (pump B overload)	Overload relay FT2 of pump motor B - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I09 / I:1/8	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	KA	BOMBA A (pump A)	Contactor KA - booster pump A
Q02 / O:2/1	KB	BOMBA B (pump B)	Contactor KB - booster pump B
Q03 / O:2/2	H1	BOMBA LIGADA (pump on)	White indicator - pump on (motor moving)
Q04 / O:2/3	H2	AUTOMÁTICO (automatic)	Blue indicator - automatic mode selected (other function)

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
Q05 / O:2/4	H3	FALTA D'ÁGUA (no water)	Red indicator - cistern at minimum level

Ready-made programs (Program menu)

- 1 - Complete: manual/automatic + pump A/B selection
- 2 - Step 1: manual start/stop of pump A
- 3 - Automatic by float switch only (with A/B selection)

Training faults (Simulation menu)

- Overload on pump motor A (overload relay FT1)
- Overload on pump motor B (overload relay FT2)
- Pump A burnt out
- Pump B burnt out
- Upper tank float stuck asking for water
- Float stuck at the top (never asks for water)
- Faulty cistern electrode (always 1)
- Cistern float valve stuck (does not close)

Scene 10 - Turntable (sorts by size)



Sensor B4 at the end of the conveyor holds the box until the table is at 0° and free. On the table, the lower sensor sees that the box has arrived and the upper one sees whether it is tall. Tall -> RIGHT, short -> LEFT.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	B1	SENSOR ENTRADA (entry sensor)	Optical sensor B1 - box at the start of the conveyor
I04 / I:1/3	B2	SENSOR SUPERIOR (upper sensor)	UPPER optical sensor B2 (on the table, at 30 cm) - only a TALL box breaks the beam
I05 / I:1/4	B3	SENSOR INFERIOR (lower sensor)	LOWER optical sensor B3 (on the table, at 6 cm) - any box that reached the position breaks the beam
I06 / I:1/5	S0	MESA EM 0° (table at 0°)	Limit switch S0 - table aligned with the conveyor (receives box)
I07 / I:1/6	SD	MESA À DIREITA (table right)	Limit switch SD - table turned +90° (unloads to the right)
I08 / I:1/7	SE	MESA À ESQUERDA (table left)	Limit switch SE - table turned 90° (unloads to the left)
I09 / I:1/8	FT1	TÉRMICO ESTEIRA (conveyor overload)	Overload relay FT1 of conveyor motor M1 - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I10 / I:1/9	FT2	TÉRMICO ROLOS (rollers overload)	Overload relay FT2 of the table roller motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I11 / I:1/10	FT3	TÉRMICO GIRO (rotation overload)	Overload relay FT3 of the table rotation motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I12 / I:1/11	B4	SENSOR FIM ESTEIRA (end-of-conveyor sensor)	Optical sensor B4 - box at the END of the conveyor, before the table (the conveyor must stop until the table is at 0° and free)
I13 / I:1/12	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch (NR-12). At rest the input is 1; when pressed it goes to 0. It only stops the machine through the program (NO contact in series).
Q01 / O:2/0	K1	ESTEIRA (conveyor)	Contactor K1 - infeed conveyor motor
Q02 / O:2/1	K2	ROLOS DA MESA (table rollers)	Contactor K2 - table roller motor (moves the box forward)
Q03 / O:2/2	K3	GIRA DIREITA (turn right)	Contactor K3 - turns the table to the right (+90°)
Q04 / O:2/3	K4	GIRA ESQUERDA (turn left)	Contactor K4 - turns the table to the left (90°)
Q05 / O:2/4	H1	LÂMPADA VERDE (green lamp)	Green indicator - system on
Q06 / O:2/5	H2	LÂMPADA BRANCA (white lamp)	White indicator - table turning (machine moving)

Ready-made programs (Program menu)

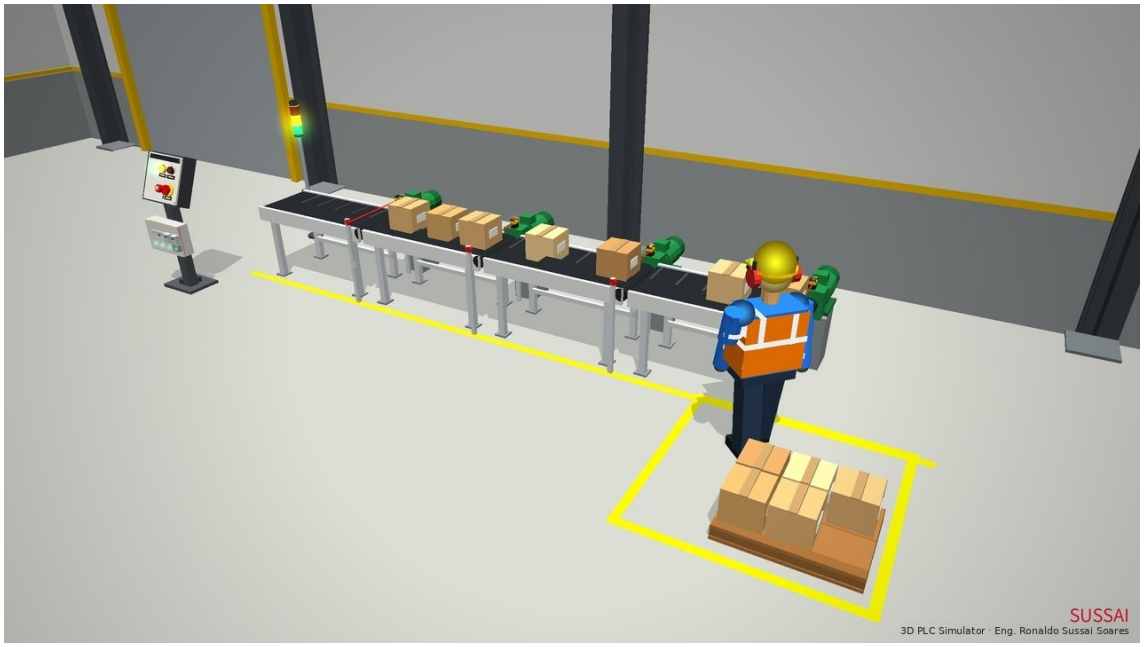
- 1 - Complete sorting (end sensor B4 + 2 sensors on the table)
- 2 - Step 1: bring the box to the center of the table

Training faults (Simulation menu)

- Overload on conveyor motor M1 (overload relay FT1)
- Overload on the table roller motor (overload relay FT2)
- Overload on the table rotation motor (overload relay FT3)
- UPPER sensor B2 with dirty lens
- Limit switch SD (right) broken
- Table roller motor burnt out

- LOWER sensor B3 stuck at 1
- Table rotation motor burnt out
- Sensor B4 (end of conveyor) with dirty lens

Scene 11 - 4 cascading conveyors (accumulation) + operator



Four conveyors in sequence; at the end an operator picks up the boxes and stacks them on the pallet. If boxes accumulate, the conveyors stop in cascade (4 -> 3 -> 2 -> 1) and restart when the operator clears the last one.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	S1	FIM ESTEIRA 1 (end conveyor 1)	Optical sensor S1 - box at the end of conveyor 1
I04 / I:1/3	S2	FIM ESTEIRA 2 (end conveyor 2)	Optical sensor S2 - box at the end of conveyor 2
I05 / I:1/4	S3	FIM ESTEIRA 3 (end conveyor 3)	Optical sensor S3 - box at the end of conveyor 3
I06 / I:1/5	S4	FIM ESTEIRA 4 (end conveyor 4)	Optical sensor S4 - box at the end of conveyor 4 (pick-up position for the operator)
I07 / I:1/6	FT1	TÉRMICO E1 (overload C1)	Overload relay FT1 of conveyor motor 1 - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I08 / I:1/7	FT2	TÉRMICO E2 (overload C2)	Overload relay FT2 of conveyor motor 2 - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I09 / I:1/8	FT3	TÉRMICO E3 (overload C3)	Overload relay FT3 of conveyor motor 3 - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I10 / I:1/9	FT4	TÉRMICO E4 (overload C4)	Overload relay FT4 of conveyor motor 4 - contact 95-96 (NC): 1 = normal, 0 = tripped by overload

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I11 / I:1/10	SE	EMERGÊNCIA (EMERGENCY)	Emergency stop button - NC contact with latch: at rest the input is 1
Q01 / O:2/0	K1	ESTEIRA 1 (conveyor 1)	Contactor K1 - conveyor motor 1
Q02 / O:2/1	K2	ESTEIRA 2 (conveyor 2)	Contactor K2 - conveyor motor 2
Q03 / O:2/2	K3	ESTEIRA 3 (conveyor 3)	Contactor K3 - conveyor motor 3
Q04 / O:2/3	K4	ESTEIRA 4 (conveyor 4)	Contactor K4 - conveyor motor 4
Q05 / O:2/4	H1	LÂMPADA VERDE (green lamp)	Green indicator - line on
Q06 / O:2/5	H2	LÂMPADA AMARELA (yellow lamp)	Yellow indicator - accumulation (some conveyor stopped waiting)
Q07 / O:2/6	H3	LÂMPADA VERMELHA (red lamp)	Red indicator - overload relay tripped

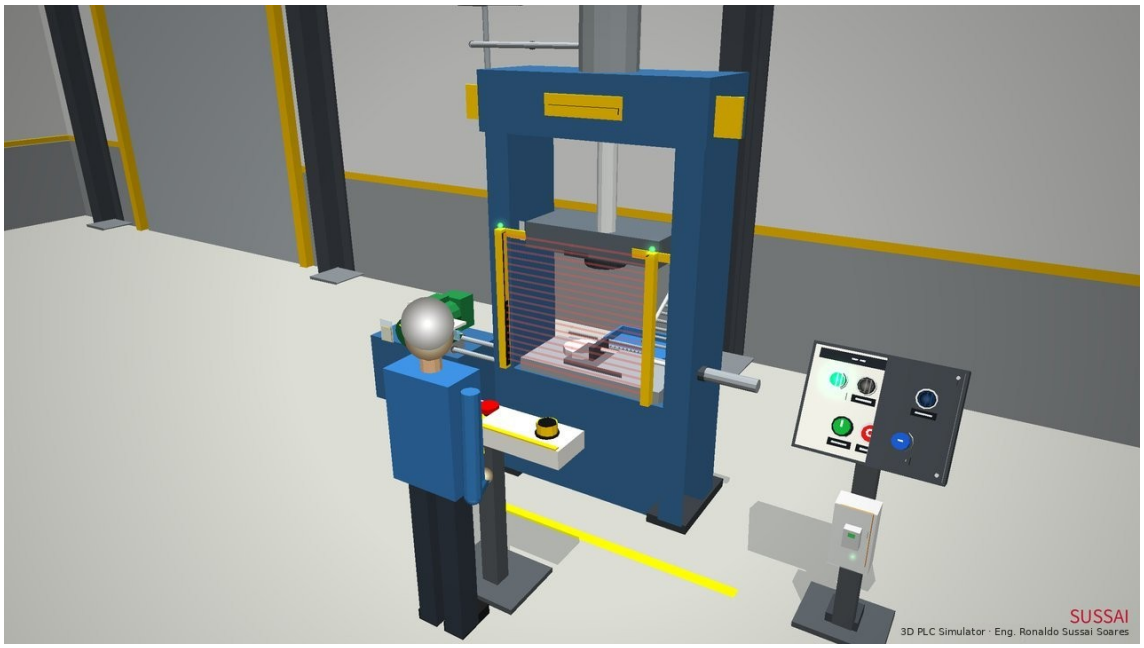
Ready-made programs (Program menu)

- 1 - Complete cascade (accumulation with end sensors)
- 2 - No cascade (intentional ERROR)
- 3 - Step 1: only conveyor 4 stops at the sensor

Training faults (Simulation menu)

- Overload on conveyor motor 1 (overload relay FT1)
- Overload on conveyor motor 2 (overload relay FT2)
- Overload on conveyor motor 3 (overload relay FT3)
- Overload on conveyor motor 4 (overload relay FT4)
- Sensor S4 (end of conveyor 4) with dirty lens
- Sensor S2 stuck at 1
- Conveyor motor 3 burnt out
- Conveyor 2 belt slipping

Scene 12 - Hydraulic press - two-hand control and light curtain (NR-12)



The press only comes down with BOTH hands on the two-hand control buttons (at the same time). The light curtain stops the press if someone puts a hand into the pressing area.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact, green, flush, symbol I (IEC 60204-1)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact, red, raised, symbol O: at rest the input is 1
I03 / I:1/2	BE	BIMANUAL ESQUERDO (left two-hand button)	LEFT two-hand button - NO, black, with protective ring
I04 / I:1/3	BD	BIMANUAL DIREITO (right two-hand button)	RIGHT two-hand button - NO, black, with protective ring
I05 / I:1/4	CL	CORTINA DE LUZ (light curtain)	Light curtain (OSSD output): 1 = area CLEAR, 0 = beam interrupted (hand/body in the pressing area)
I06 / I:1/5	S2	PRENSA EM CIMA (press up)	Limit switch S2 - ram retracted (up)
I07 / I:1/6	S3	PRENSA EMBAIXO (press down)	Limit switch S3 - ram at the end of the stroke (part pressed)
I08 / I:1/7	SE	EMERGÊNCIA (EMERGENCY)	Emergency stop button - red mushroom on a yellow background, NC with latch: at rest the input is 1
I09 / I:1/8	FT1	TÉRMICO BOMBA (pump overload)	Overload relay FT1 of the hydraulic pump motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I10 / I:1/9	S4	REARME (reset)	RESET button - NO (blue): resets the safety after the light curtain or the emergency stop has acted
Q01 / O:2/0	K1	BOMBA HIDRÁULICA (hydraulic pump)	Contactor K1 - hydraulic power unit pump motor
Q02 / O:2/1	Y1	DESCE (down)	Solenoid Y1 - directional valve: ram moves DOWN
Q03 / O:2/2	Y2	SOBE (up)	Solenoid Y2 - directional valve: ram moves UP
Q04 / O:2/3	H1	LÂMPADA VERDE (green lamp)	Green indicator - normal condition: hydraulic unit on, ready

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
Q05 / O:2/4	H2	LÂMPADA BRANCA (white lamp)	White indicator - ram moving (machine moving)
Q06 / O:2/5	H3	LÂMPADA VERMELHA (red lamp)	Red indicator - protective stop: emergency, curtain interrupted or overload
Q07 / O:2/6	H4	LÂMPADA AZUL (blue lamp)	Blue indicator - operator action: press RESET

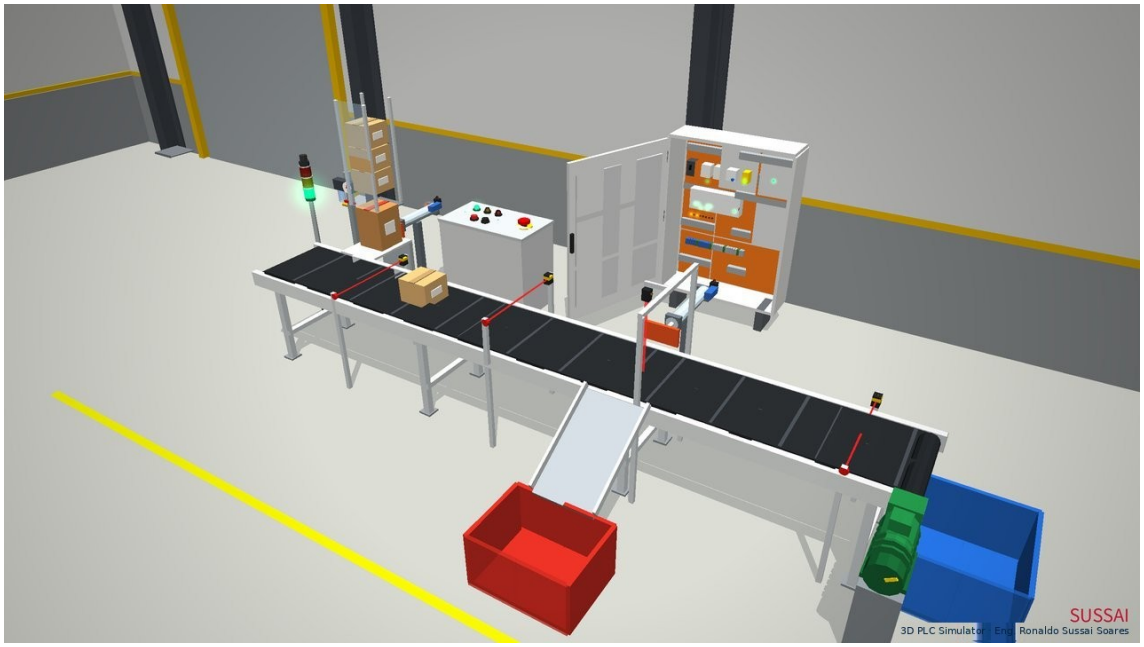
Ready-made programs (Program menu)

- 1 - Two-hand control with simultaneity + light curtain (complete)
- 2 - With an ERROR: two-hand control without simultaneity
- 3 - With an ERROR: the light curtain is not in the program

Training faults (Simulation menu)

- Overload on the hydraulic pump motor (overload relay FT1)
- LEFT two-hand button tied down (operator bypass)
- Dirty / misaligned light curtain
- Faulty limit switch S3 (down)

Scene 13 - Conveyor with sorting (advanced)



Complete line: pneumatic feeder, sensors, diverter, electrical panel, NR-12 emergency stop and overload relay.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	S1	LIGA (START)	START button - NO contact (green)
I02 / I:1/1	S0	DESLIGA (STOP)	STOP button - NC contact (red)
I03 / I:1/2	SE	EMERGÊNCIA (EMERGENCY)	Emergency stop button - NC with latch (NR-12)

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I04 / I:1/3	S2	REVERSO (reverse)	REVERSE button - NO contact, YELLOW (reverse direction - IEC 60204-1)
I05 / I:1/4	S3	REARME (reset)	RESET button - NO, BLUE (a function that is not start, stop or reverse)
I06 / I:1/5	SS	SELETORA (selector)	Selector switch MAN (0) / AUTO (1)
I07 / I:1/6	F7	TÉRMICO (overload)	Overload relay F7 - contact 95-96 (NC)
I08 / I:1/7	B1	SENSOR INÍCIO (start sensor)	Optical sensor B1 - start of the conveyor
I09 / I:1/8	B2	SENSOR DESVIO (diverter sensor)	Diffuse optical sensor B2 - in front of the diverter
I10 / I:1/9	B3	SENSOR FIM (end sensor)	Optical sensor B3 - end of the conveyor
I11 / I:1/10	B4	SENSOR ALTURA (height sensor)	Optical sensor B4 - detects TALL boxes only
I12 / I:1/11	SA0	ALIM. RECUADO (feeder retracted)	Magnetic sensor - feeder cylinder A retracted
I13 / I:1/12	SA1	ALIM. AVANÇADO (feeder extended)	Magnetic sensor - feeder cylinder A extended
I14 / I:1/13	SB0	DESV. RECUADO (diverter retracted)	Magnetic sensor - diverter cylinder B retracted
I15 / I:1/14	SB1	DESV. AVANÇADO (diverter extended)	Magnetic sensor - diverter cylinder B extended
Q01 / O:2/0	K1	MOTOR AVANÇO (motor forward)	Contactor K1 - motor M1 forward direction
Q02 / O:2/1	K2	MOTOR REVERSO (motor reverse)	Contactor K2 - motor M1 reverse direction
Q03 / O:2/2	H1	SINAL VERDE (green signal)	Green indicator - machine ready to operate / in normal operation
Q04 / O:2/3	H2	SINAL AMARELO (yellow signal)	Yellow indicator - attention (warning before start, batch complete)
Q05 / O:2/4	H3	SINAL VERMELHO (red signal)	Red indicator - abnormal condition: fault / protective stop
Q06 / O:2/5	Y1	ALIMENTADOR (feeder)	Solenoid Y1 - cylinder A (single-acting, spring return)
Q07 / O:2/6	Y2	DESVIADOR (diverter)	Solenoid Y2 - cylinder B (single-acting, spring return)
Q08 / O:2/7	HA1	SIRENE (siren)	Warning siren / buzzer

Ready-made programs (Program menu)

- 1 - Direct-on-line start with seal-in
- 2 - Reversing with interlock
- 3 - Timed start with warning (NR-12)
- 4 - Complete automatic conveyor

Training faults (Simulation menu)

- START button (S1) with damaged NO contact
- STOP button (S0) wire broken at the terminal
- Sensor B1 stuck at level 1
- Sensor B3 with dirty lens (does not detect)

- Height sensor B4 misaligned
- Contactor K1 coil burnt out
- K1 power contacts welded (stuck)
- PLC output Q5 damaged
- No compressed air in the line
- Mechanical overload on motor M1
- Phase loss on motor M1
- Conveyor belt slipping
- Indicator lamp H1 (green) burnt out
- Feeder magnetic sensor SA1 out of adjustment

Scene 14 - Passenger elevator (4 floors)



Elevator with ground floor + 3 floors: registered calls, floor sensors, automatic door with light curtain, traction machine with overload relay. Press a call button: YOU appear on the floor, enter the car and choose which floor to go to.

Inputs and outputs

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I01 / I:1/0	BT	CHAMADA TÉRREO (ground floor call)	Ground floor call button (landing and car) - NO
I02 / I:1/1	B1	CHAMADA 1º ANDAR (1st floor call)	1st floor call button (landing and car) - NO
I03 / I:1/2	B2	CHAMADA 2º ANDAR (2nd floor call)	2nd floor call button (landing and car) - NO
I04 / I:1/3	B3	CHAMADA 3º ANDAR (3rd floor call)	3rd floor call button (landing and car) - NO
I05 / I:1/4	S0	SENSOR TÉRREO (ground floor sensor)	Floor sensor S0 - car level at the ground floor

Address (Clic02 / LogixPro)	Tag	Name (as shown in the program)	Description
I06 / I:1/5	S1	SENSOR 1º ANDAR (1st floor sensor)	Floor sensor S1 - car level at the 1st floor
I07 / I:1/6	S2	SENSOR 2º ANDAR (2nd floor sensor)	Floor sensor S2 - car level at the 2nd floor
I08 / I:1/7	S3	SENSOR 3º ANDAR (3rd floor sensor)	Floor sensor S3 - car level at the 3rd floor
I09 / I:1/8	FCF	PORTA FECHADA (door closed)	Limit switch: door fully closed (and locked)
I10 / I:1/9	FCA	PORTA ABERTA (door open)	Limit switch: door fully open
I11 / I:1/10	CL	CORTINA DE LUZ (light curtain)	Door light curtain: 1 = person/object in the doorway
I12 / I:1/11	FT1	TÉRMICO MÁQUINA (machine overload)	Overload relay FT1 of the traction machine motor - contact 95-96 (NC): 1 = normal, 0 = tripped by overload
I13 / I:1/12	EM	EMERGÊNCIA (EMERGENCY)	Emergency stop button - NC contact: at rest the input is 1
Q01 / O:2/0	K1	SOBE (up)	Contactor K1 - traction machine motor UP direction
Q02 / O:2/1	K2	DESCE (down)	Contactor K2 - traction machine motor DOWN direction
Q03 / O:2/2	K3	ABRE PORTA (open door)	Contactor K3 - door operator OPEN
Q04 / O:2/3	K4	FECHA PORTA (close door)	Contactor K4 - door operator CLOSE
Q05 / O:2/4	HT	REGISTRO TÉRREO (ground floor registered)	Registered call lamp - ground floor (blue)
Q06 / O:2/5	H1	REGISTRO 1º (1st registered)	Registered call lamp - 1st floor (blue)
Q07 / O:2/6	H2	REGISTRO 2º (2nd registered)	Registered call lamp - 2nd floor (blue)
Q08 / O:2/7	H3	REGISTRO 3º (3rd registered)	Registered call lamp - 3rd floor (blue)

Ready-made programs (Program menu)

- 1 - Complete elevator (calls, door and direction)
- 2 - Step 1: go up and down between ground floor and 3rd floor

Training faults (Simulation menu)

- Overload on the traction machine motor (overload relay FT1)
- Faulty floor sensor S2
- Dirty light curtain (always obstructed)
- Jammed door operator

9. Gamification Mode

Each exercise opens the machine with a ready program and a panel with the steps. Each step is of one type:

Type	What the student does
Do it on the machine	Performs the action on the machine (e.g.: start with seal-in). SUSSAI checks it on the machine itself.
Pergunta (Question)	Answers a technical question about what happened.
Find the fault	A fault is hidden; the student observes, measures in the I/O panel and points out the cause.
Corrija o programa (Fix the program)	The program has an intentional error; the student fixes the Ladder.

Each correct answer adds points; mistakes and hints take points away. At the end the **score from 0 to 10** appears with stars, and it is kept in **Meus resultados** (My results, per user). In SUSSAI Teste, only the exercises of scenes 01 to 03 are available.

Available exercises

Nº	Exercise	Machine	Level	Steps / points
E01	**Direct-on-line start with seal-in and overload relay** Start, stop, seal-in, motor overload, overload reset and a hidden fault.	01 - Simple conveyor (sensor + box)	Easy	7 steps / 100 pts
E02	**Fan: overload and broken wire on STOP** Test the direct-on-line start and the overload relay, and see why the stop button is NC.	02 - Industrial fan (direct-on-line start)	Easy	8 steps / 100 pts
E03	**Missing START button - unexpected start-up (NR-12)** The program has no START button contact. Find the problem and fix the diagram.	02 - Industrial fan (direct-on-line start)	Medium	5 steps / 100 pts
E04	**Build it yourself: fan direct-on-line start** Blank diagram: build the direct-on-line start with seal-in, indicators and overload relay and complete the 6 missions.	02 - Industrial fan (direct-on-line start)	Hard	6 steps / 100 pts
E05	**Level control and burnt-out sensor** Automatic filling with low and high level sensors, consumption and a fault that makes the tank overflow.	03 - Mixing tank (pump + agitator)	Easy	6 steps / 100 pts
E06	**Traffic light with pedestrian crossing** On-demand light on a two-way street: it only closes when a pedestrian asks. Timing, minimum green, all-red and a push-button fault.	04 - Single-road traffic light (two-way) with pedestrian crossing	Easy	6 steps / 100 pts
E07	**Turntable: sorting by size** Test the sorting of small and large boxes and find the faulty sensor.	10 - Turntable (sorts by size)	Medium	6 steps / 100 pts
E08	**Cascading conveyors: accumulation** Stop the operator, watch the conveyors stop in cascade and restart; find the faulty motor.	11 - 4 cascading conveyors (accumulation) + operator	Medium	6 steps / 100 pts
E09	**Filling: overflowing bottle** Automatic filling with position and full-bottle sensors; find out why the bottle overflows.	08 - Bottle filling (silo + valve)	Medium	4 steps / 100 pts
E10	**Elevator: calls, door and floor sensor** Call the elevator, understand the door safety and find the faulty floor sensor.	14 - Passenger elevator (4 floors)	Hard	4 steps / 100 pts
E11	**Reversing without interlock (short circuit)** The forward/reverse program has no interlock. Cause the short circuit, understand it and fix it.	13 - Conveyor with sorting (advanced)	Hard	7 steps / 100 pts

Nº	Exercise	Machine	Level	Steps / points
E12	**Hydraulic press: two-hand control and light curtain (NR-12)** Two-hand control with simultaneity, light curtain, reset and a hidden bypass fault.	12 - Hydraulic press - two-hand control and light curtain (NR-12)	Medium	8 steps / 100 pts

10. Extra scenes (.cena files)

Anyone can create a new machine and distribute it as a file with the **.cena** extension (example: Portao_automatico.cena).

- **Install:** menu Scene > "Add new scene (.cena)..." and choose the file, or drag the .cena file into the window. The scene is added at the end of the Scene menu and is saved on the computer. Installing a file with the same "id" again updates the scene.
- **Automatic test:** before installing, SUSSAI tests the scene; if there is an error, it shows the reason and the line of the file, and nothing is installed.
- **Create:** Scene > Extra scenes > "Download the TEMPLATE" (a commented sliding gate example). Edit it in Notepad or VS Code, change "id" and "name" and install it to test.
- **Distribute / remove:** Scene > Extra scenes > download the .cena of an installed scene; the trash can removes it.
- The emergency stop button (NR-12) is added automatically if the scene does not have one.

Attention: a .cena file contains program code. Install only scenes from trusted sources. Extra scenes require SUSSAI Completo.

11. Safety and applied standards

Button and indicator colors (IEC 60204-1 / IEC 60073)

Element	Color	Meaning
Button	Green or black	Start / switch on
Button	Red	Stop, switch off and emergency
Button	Yellow	Reverse direction / cancel abnormal condition
Button	Blue or white	Other functions (e.g.: reset)
Indicator	Red	Abnormal condition / danger
Indicator	Yellow	Attention
Indicator	Green	Ready to operate
Indicator	White	In operation (moving)
Indicator	Blue	Other functions

Emergency stop button (NR-12)

Red mushroom on a yellow background, NC contact with latch: at rest the input is 1 and, when pressed, it goes to 0. In SUSSAI the emergency stop is **only a PLC input**: no output, lamp or motor responds to it by itself. The **program** is what stops the machine (NO emergency contact in series with STOP - the ready-made programs already include it). Pressing it latches; to release, click again. With the seal-in in the program, the machine **does not restart by itself**: a new command is needed.

Two-hand control and light curtain (scene 12 - Press)

The press only comes down with **both hands** on the buttons at the same time (simultaneity within 0.5 s, following the principles of ISO 13851). The light curtain stops the press if something enters the danger zone. SUSSAI also simulates the **bypass** (button tied down), so the student can recognize it and understand why it is forbidden.

Programming language

SUSSAI Ladder follows the concepts of IEC 61131-3 (contacts, coils, TON/TOF timers and CTU/CTD counters), with WEG Clic02 or Allen-Bradley LogixPro style naming.

> SUSSAI is an educational simulator. It does not replace design, risk assessment, engineering responsibility and the standards applicable to real machines.

12. Keyboard shortcuts

Key	Action
F5 / F6	PLC RUN / STOP
Ctrl+S	Save to My circuits
Ctrl+N	New blank program
Ctrl+Z / Ctrl+Y	Undo / redo
Delete	Delete the selected element
1 2 3 6 7 8 9 0	Diagram tools
Esc	Close the menu / Select tool
Ctrl+Shift+G (entry screen)	Record the cover video and scene photos (for promotion)

13. Saving, opening and moving to another computer

- **File > Save (Ctrl+S):** stores it in "My circuits", on the computer itself.
- **File > Download .json file:** creates a circuit file to take to another computer or hand in to the instructor.
- **File > Open file from the computer (.json):** loads a saved circuit. If the file belongs to another machine, SUSSAI switches scenes automatically.
- **File > Export the diagram as an image (PNG):** saves an image of the diagram for reports.

14. Troubleshooting

Problem	Solution
SUSSAI does not open and shows "Windows protected your PC".	Click "More info" > "Run anyway".
The Google window does not appear.	Allow pop-ups/windows for SUSSAI and try again. Check the internet connection.
"Could not confirm your Google account".	Sign in again with the correct account. It must be the same account used for the purchase.
"Limit of computers reached".	On the old computer: Help > SUSSAI license > Disconnect this computer. Or ask for release by e-mail.
"Incorrect computer date".	Correct the Windows date and time and click Try again.
The free trial has ended.	Buy a plan with the "I want to buy" button or by e-mail. Your circuits remain saved.
I bought it, but it is still in trial.	Click "I already bought – check now". In SUSSAI Teste, use the "Download SUSSAI Completo" button.
SUSSAI Completo says it needs a paid license.	It does not open with the free trial. Use SUSSAI Teste until the purchase is released.
The 3D image is slow or stuttering.	View > 3D image quality > Low. Close other heavy programs.
The machine does not start even with the correct program.	Check: PLC in RUN (F5)? Emergency stop latched? Overload relay tripped? Fault active in the Simulation menu?

15. Early Access: reporting problems and suggestions

SUSSAI version 1.0 is in **Early Access** (testing phase). It is already complete and works, but it is being tested on many different computers, so some detail may need fixing. The yellow **ACESSO ANTECIPADO** (EARLY ACCESS) badge appears on the entry screen and at the top of the program; click it to read the notice. The notice also appears by itself once for each new version.

How to report a problem (takes 1 minute)

- Menu **Help > Report a problem (bug)** (or click the EARLY ACCESS badge, or the link on the welcome screen).
- Write **what happened**. If you know, explain **how to make it happen again** (step by step) and **what you expected**.
- Fill in name, WhatsApp and city/school (optional; SUSSAI remembers them for next time).
- Keep the option **Include the technical information** checked: version, build, scene, mode, PLC style, Windows, screen size and the latest program errors. No password or personal file is sent.
- Click **Open in Gmail** (the message already has a ready subject and text, to contato.sussai@gmail.com) or **Other e-mail program**. No e-mail on the computer? Use **Copy the message** and paste it anywhere (even on WhatsApp).
- **Tip:** a screenshot helps a lot. Press Win+Shift+S, cut out the screen and paste it (Ctrl+V) into the e-mail.

Suggestions, questions and compliments

Use **Help > Send a suggestion** or **Help > Question or compliment to the creator**. Ideas for new machines and exercises are very welcome.

What's new and new versions

Help > What's new in this version shows what changed. The **build** (e.g.: 2026.10.19) appears in Help > About SUSSAI. When a new version with fixes is released, SUSSAI shows a notice (at most once a day, with internet): click it to download from the website. **Your license remains valid** in the new version.

16. Contact and rights

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Support, purchases and renewals: **contato.sussai@gmail.com** - **https://sussai-software.github.io**

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